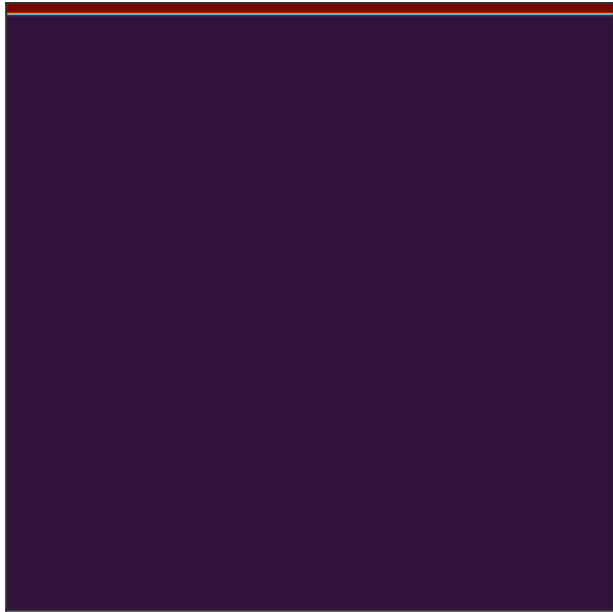


Spiral Wave Initiation via S1-S2 Protocol
Barkley Reaction-Diffusion Model (2D Isotropic Domain)

$N=256, dx=0.5, D=1.0, a=0.8, b=0.01, s=0.02$
 $\tau=256, \sigma=1.0$
S1: Planar stimulus (top strip) → S2: Premature cross-field stimulus in repolarization wake → Wave breaks and curls into self-sustaining rotor (reentry). Parameters chosen for stable spiral formation. Generated: 2026-05-19 08:45

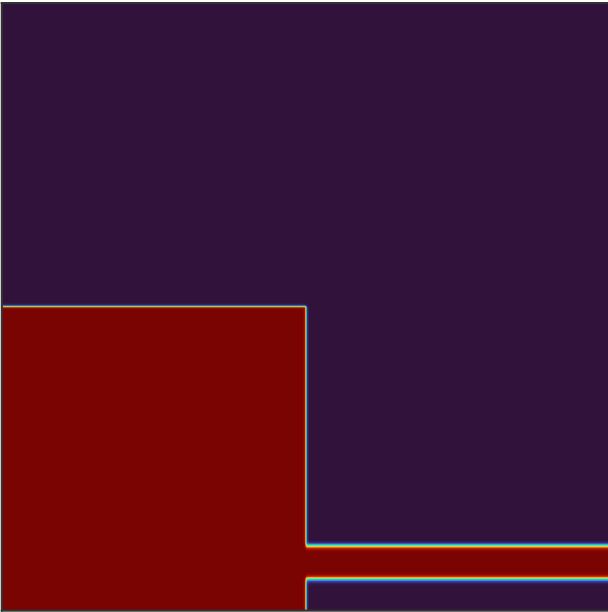
t = 0 ms
S1 Planar Wave Initiated



t = 20 ms
S1 Propagating



t = 28 ms
S2 Premature Stimulus Applied
(Wavefront Broken)



t = 80 ms
Wavefront Begins to Curl



t = 150 ms
Spiral Forming
(Phase Singularity)



t = 250 ms
Mature Rotor
(Reentrant Wave)



t = 400 ms
Stable Spiral Rotation



t = 550 ms
Sustained Spiral Wave

